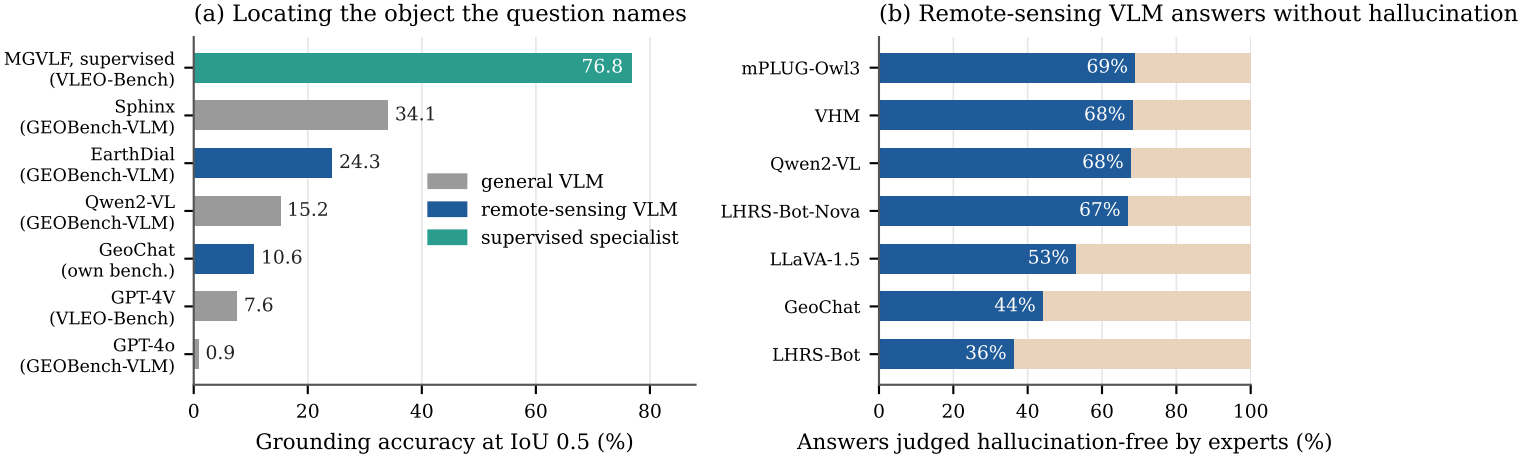
(a) Locating the object the question names
MGVLF, supervised (VLEO-Bench)
Sphinx (GEOBench-VLM)
EarthDial (GEOBench-VLM)
Qwen2-VL (GEOBench-VLM)
GeoChat (own bench.)
GPT-4V (VLEO-Bench)
GPT-4o (GEOBench-VLM)
76.8
34.1
24.3
15.2
10.6
7.6
0.9
general VLM
remote-sensing VLM
supervised specialist
0
20
40
60
80
Grounding accuracy at IoU 0.5 (%)
(b) Remote-sensing VLM answers without hallucination
mPLUG-Owl3
VHM
Qwen2-VL
LHRS-Bot-Nova
LLaVA-1.5
GeoChat
LHRS-Bot
69%
68%
68%
67%
53%
44%
36%
0
20
40
60
80
100
Answers judged hallucination-free by experts (%)